Diagonalization and the Minimal Polynomial: Answer Key

Problem sheet for the Day 7 class · CSIR NET previous year questions

Answers

Q	1	2	3	4	5	6	7	8	9	10
Answer	(A)	(D)	(B)	(A),(C)	(A),(B),(D)	(C),(D)	(D)	(A),(D)	(C),(D)	(A)

Hints and short solutions

1. **(A)** CSIR NET June 2016 Cayley–Hamilton: $p(B) = 0$, so $e^{p(B)} = e^0 = I$.
2. **(D)** CSIR NET Dec 2015 The fifth eigenvalue is $15 - (2 + 2 + 3 + 3) = 5$, so $\det A = 2^2 \cdot 3^2 \cdot 5 = 180$.
3. **(B)** CSIR NET Dec 2018 $A = UDU^*$ with D real, so the eigenvalues are $\frac{\lambda - i}{\lambda + i}$ with $\lambda \in \mathbb{R}$, and $|\lambda - i| = |\lambda + i|$. (This is the Cayley transform; it is unitary.)
4. **(A),(C)** CSIR NET June 2017 (A) distinct eigenvalues 1, 4, 6. (B) eigenvalues $\pm i, 1$: not over $\mathbb{R}$. (C) real symmetric. (D) non-zero nilpotent, never diagonalizable.
5. **(A),(B),(D)** CSIR NET June 2017 Nullity of A is 1, so 0 is an eigenvalue. Geometric multiplicities $1 + 2 + 1 = 4$, so $A \sim \mathrm{diag}(0, 2, 2, 4)$: $\det A = 0$, $\mathrm{tr}\, A = 8$, and A is diagonalizable.
6. **(C),(D)** CSIR NET June 2018 $\chi_T = (x - 1)^n$; Cayley–Hamilton gives $(T - I)^n = 0$, hence also $(T - I)^{2n} = 0$. $T = J_n(1)$ (one Jordan block) refutes (A) and (B).
7. **(D)** CSIR NET June 2018 The minimal polynomial divides $x^8 - 1$. I (degree 1), $\mathrm{diag}(1, 1, -1)$ (degree 2) and $\mathrm{diag}(R_{\pi/4}, 1)$ (degree 3) all work, and so does every conjugate SRS^{-1}: uncountably many.
8. **(A),(D)** CSIR NET Dec 2018 $T(I - T) = I$, so T is invertible; $T - I = T^2$ is invertible. The minimal polynomial divides $x^2 - x + 1$, whose roots are non-real. $T^3 = T \cdot T^2 = T^2 - T = -I$.
9. **(C),(D)** CSIR NET Dec 2018 A^2 has eigenvalues $1, 1, 1$. Two blocks means $J_2(1) \oplus J_1(1)$, so $(A - I)^2 = 0$, which is diagonal. $J_3(1)$ refutes (A) and (B). (The paper prints $(A - 1)^2$ for $(A - I)^2$.)
10. **(A)** CSIR NET Dec 2016 A is the cyclic permutation matrix: $\chi_A = x^n - 1$, whose roots are the n distinct n th roots of unity. For odd n the only real one is 1.